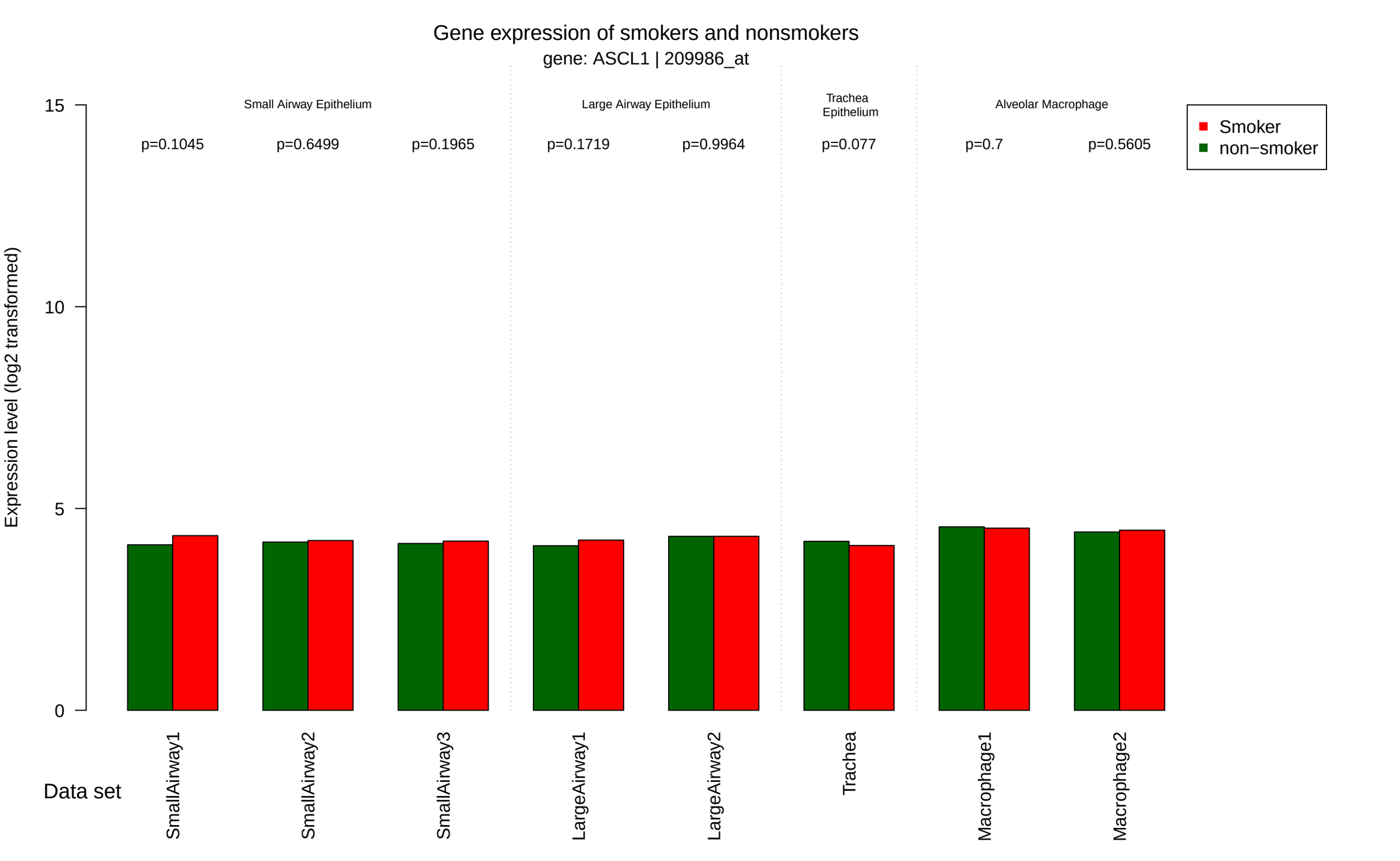
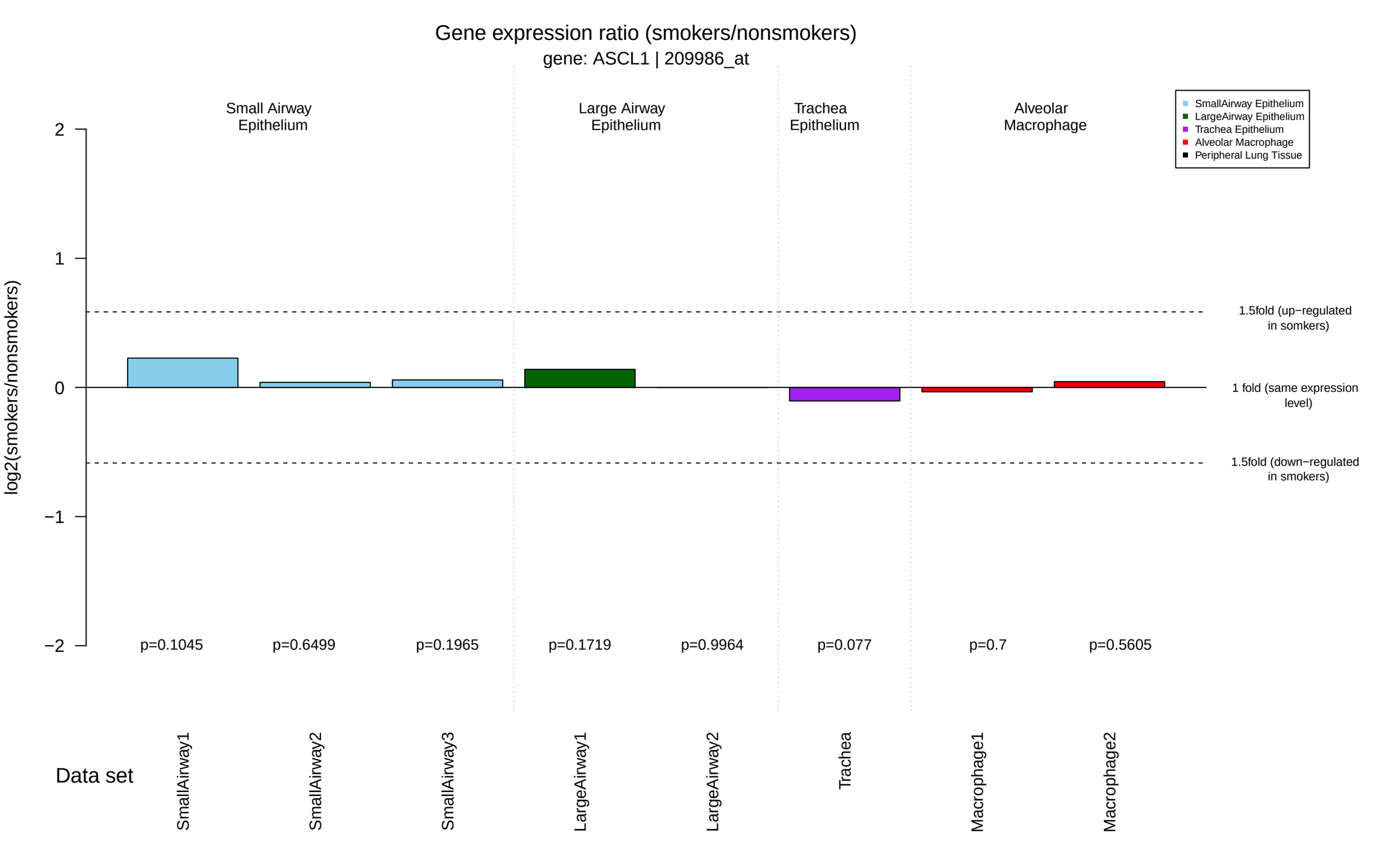
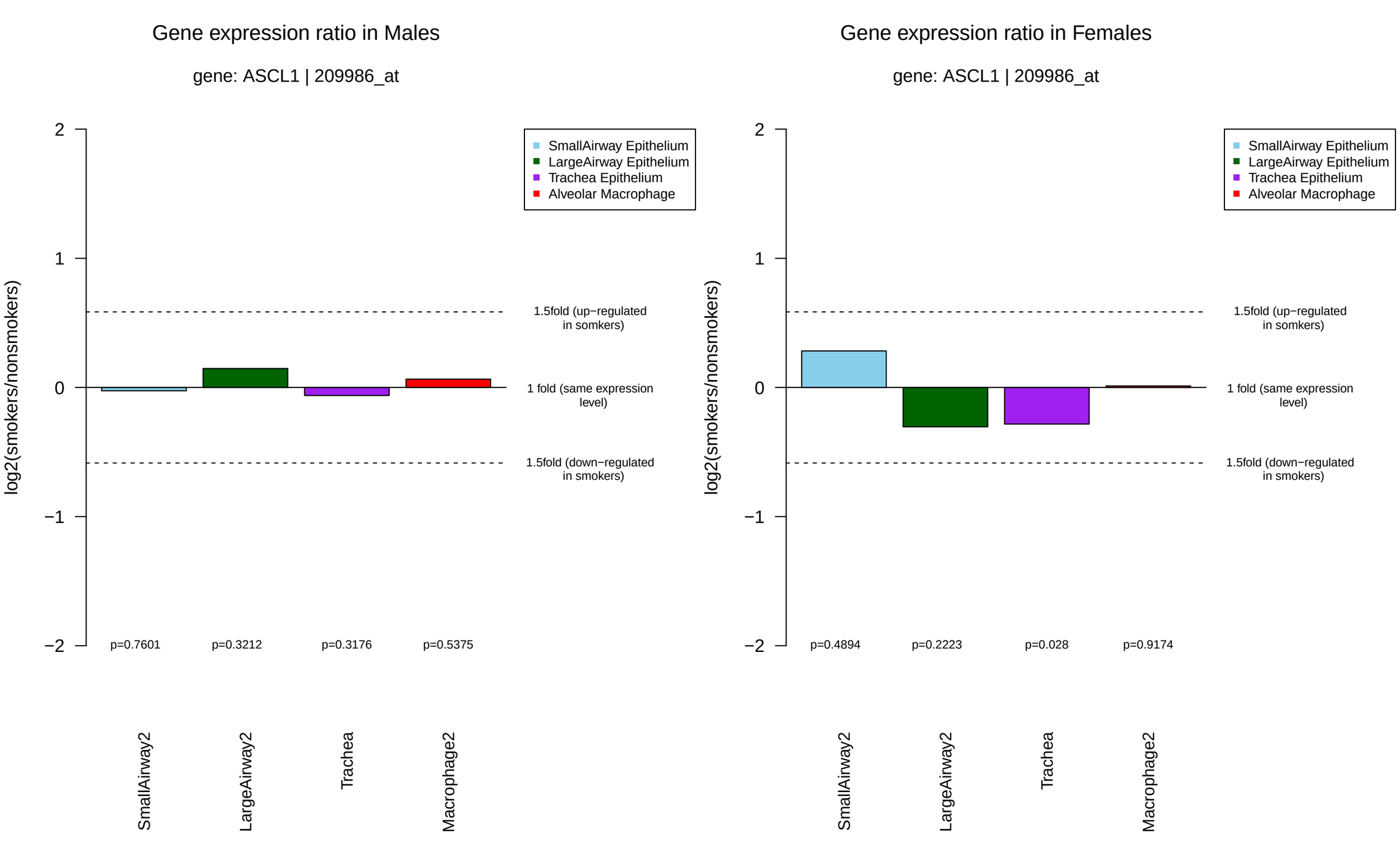
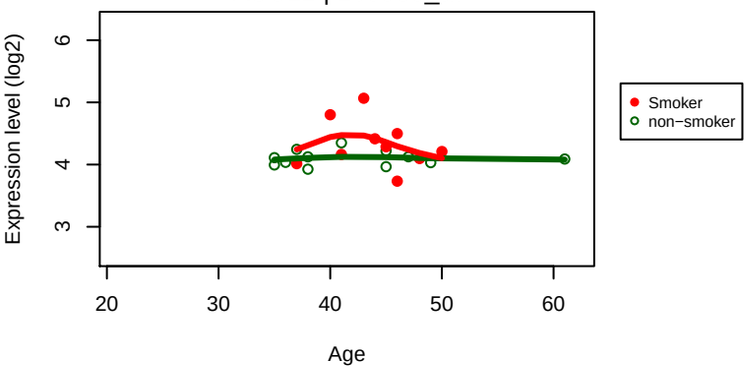
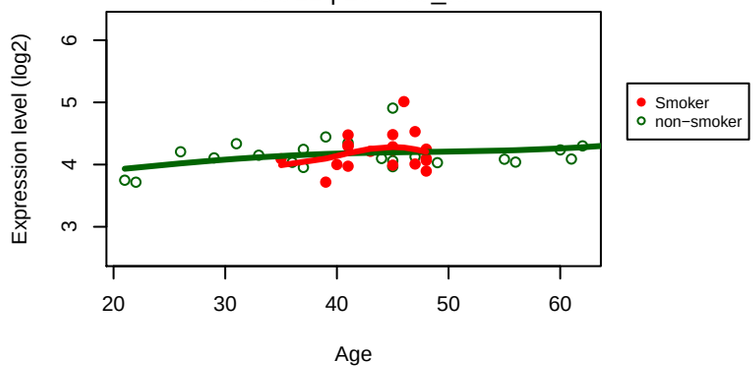
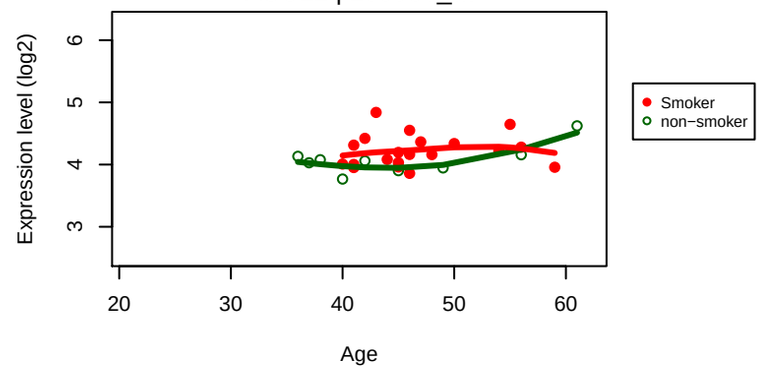

Gene expression of smokers and nonsmokers
gene: ASCL1 | 209986_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1045
p=0.6499
p=0.1965
p=0.1719
p=0.9964
p=0.077
p=0.7
p=0.5605
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

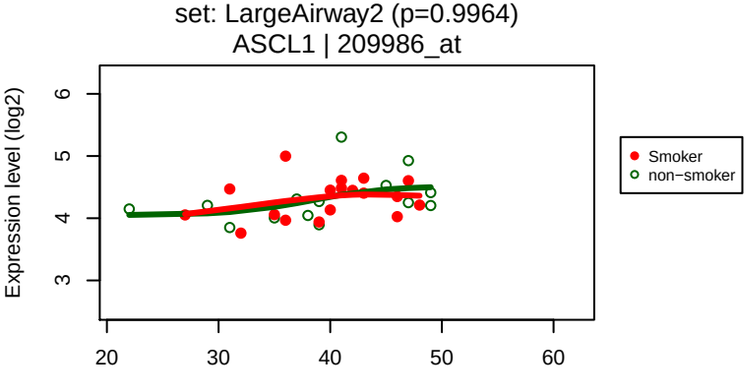

Gene expression ratio (smokers/nonsmokers)
gene: ASCL1 | 209986_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1045
p=0.6499
p=0.1965
p=0.1719
p=0.9964
p=0.077
p=0.7
p=0.5605
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

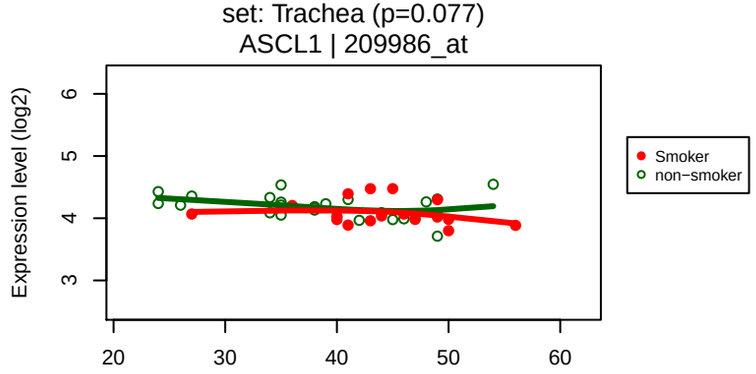

Gene expression ratio in Males
gene: ASCL1 | 209986_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7601
p=0.3212
p=0.3176
p=0.5375
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASCL1 | 209986_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4894
p=0.2223
p=0.028
p=0.9174
SmallAirway2
LargeAirway2
Trachea
Macrophage2

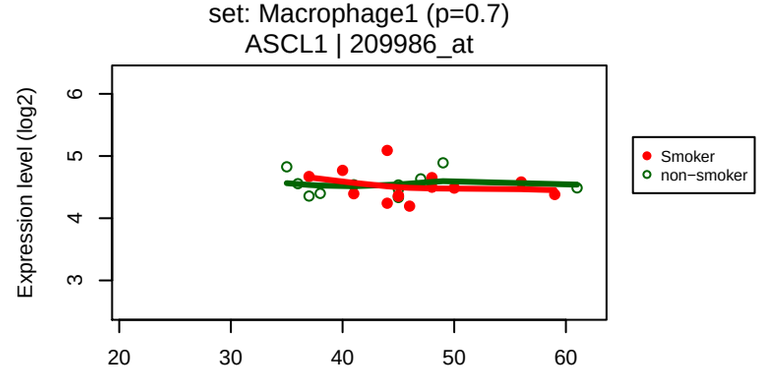

set: SmallAirway1 (p=0.1045)
ASCL1 | 209986_at
Expression level (log2)
Age
Smoker
non−smoker

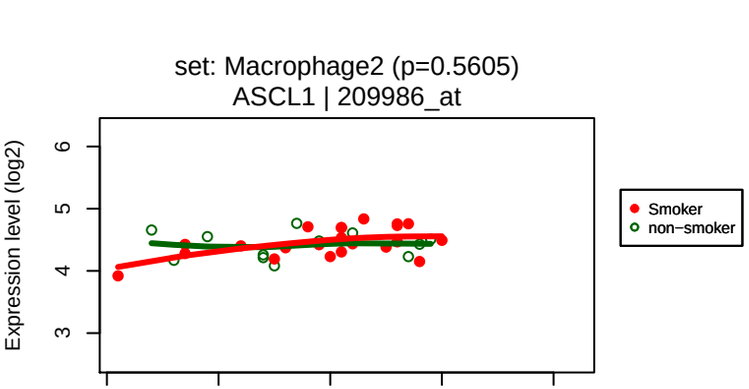

set: SmallAirway2 (p=0.6499)
ASCL1 | 209986_at
Expression level (log2)
Age
Smoker
non−smoker

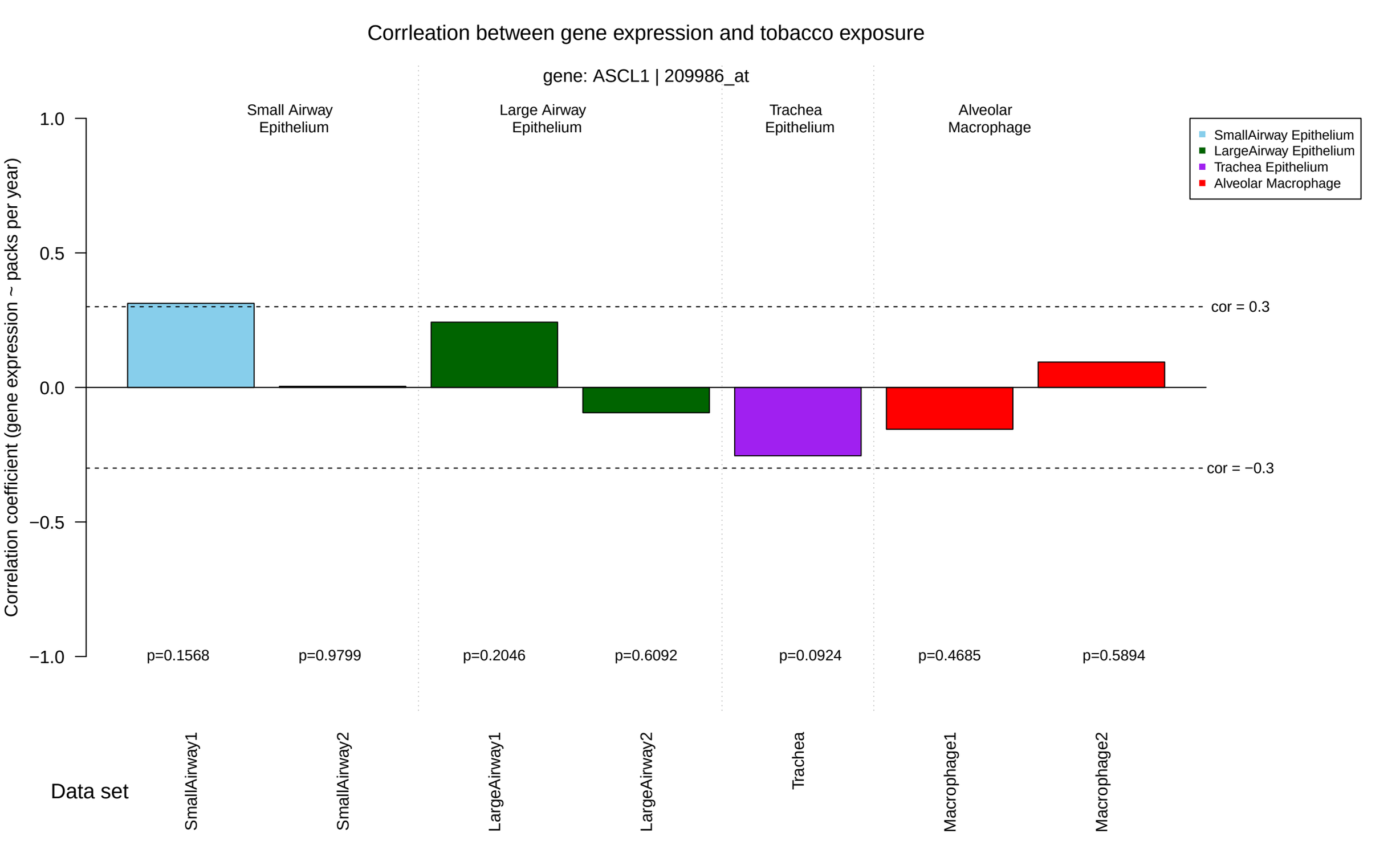

set: LargeAirway1 (p=0.1719)
ASCL1 | 209986_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9964)
ASCL1 | 209986_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.077)
ASCL1 | 209986_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7)
ASCL1 | 209986_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5605)
ASCL1 | 209986_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASCL1 | 209986_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1568
p=0.9799
p=0.2046
p=0.6092
p=0.0924
p=0.4685
p=0.5894
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2